Where Filament Segmentation Fails: Model Errors Concentrate on Filaments the Annotators Disagree On

A Solution to the Solar Filament Segmentation Challenge 2026

Saifuddin Abdullah Rasyid
Indonesia
syehsaifuddin@gmail.com
syehsyoh@yahoo.com

ABSTRACT

This report describes our solution to the Solar Filament Segmentation Challenge 2026, a Kaggle competition on automatic segmentation of solar filaments in GONG H-α observations. Our pipeline detects filaments with two YOLO11s detectors running at 1280 and 1920 px, fuses their boxes, turns each box into a mask with a box-seeded ResNet34 U-Net, and assembles strictly non-overlapping instances. It reaches a Panoptic Quality (PQ) of 0.444 (95% CI 0.419–0.467) on held-out training observations and 0.39 on the public leaderboard, and the repository reproduces the submitted file byte-for-byte from the included weights. Our main finding concerns the labels rather than the model: on held-out observations with several annotators, the model finds 84% of the filaments that every annotator marked but only 34% of those marked by a single annotator, and 53% of its "false positives" with respect to one annotator were drawn by another. Judged against the same reference annotators, the model scores PQ 0.45 while an annotator scores 0.34 (difference +0.11, 95% CI +0.08 to +0.14). Six further detector and refiner variants all landed between 0.42 and 0.45, consistent with label ambiguity, not model capacity, limiting the score.

1 INTRODUCTION

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[1]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1].

We contribute (i) a two-stage detect-then-refine pipeline whose every reported number is reproducible from a public repository[2] (code, weights, notebook), and (ii) an error analysis on held-out training observations showing that, on observations with several annotators, most remaining errors fall on filaments whose very existence the expert annotators disagree on. Inter-rater variability is a known limit of segmentation benchmarks [8, 13]; we quantify it for MAGFiLO at the instance level, with the competition's own metric. We report negative results as well, because they are what support the second point.

2 METHODOLOGY

2.1 Pipeline

Figure 1 shows the pipeline; all settings are in one configuration file of the repository.

Pre-processing. Each observation is read as a 2048×2048 grayscale image. No other pre-processing is applied to the full image; contrast enhancement happens per crop in Stage 2.

Stage 1: detection. Two YOLO11s detectors [6] (COCO-pretrained, fine-tuned on filament boxes) run at input sizes 1280 and 1920 px and keep boxes with confidence $\geq$0.05 (NMS IoU 0.5). Their boxes are merged by weighted box fusion [11]: boxes are visited by decreasing confidence; a box joins the best-overlapping cluster (IoU >0.55) that holds no box of the same detector, the cluster box is the confidence-weighted mean of its members, and its score is the sum of member confidences divided by the number of detectors, so a box seen by only one detector is down-weighted.

Stage 2: box-seeded refinement. For each fused box we take a square window of 1.5× the longer box side (at least 96 px), shifted to stay inside the image, and resize it to 512×512. A U-Net [9] with an ImageNet ResNet34 encoder [4, 5] receives three channels: the crop, its CLAHE-enhanced copy [15] (clip 2.0, 8×8 tiles), and a filled box seed padded by 60% of the box size, which tells the network *which* filament in the crop to segment. Probabilities are averaged over the identity, horizontal-flip and vertical-flip inputs, upsampled to native resolution with linear interpolation, thresholded at 0.5, and reduced to the largest connected component.

Stage 3: instance assembly. Masks are visited by decreasing fused score down to 0.32. Pixels already claimed by a stronger instance are removed, so the output never contains overlapping instances, and remnants under 200 px are dropped. An observation with no box above 0.32 keeps its single best candidate, if it has one.

Efficiency. The 180 test observations take 266 s on one Tesla T4 (1.5 s per observation, all three models, TTA included); the three weight files total 130 MB.

2.2 Training data, sampling and splits

We use only the competition training set: 707 observations with one or more annotator passes each, and only their segmentation masks. The detector targets are the boxes stored with each annotation, which coincide with the tight boxes of the masks (99.2%

[1] www.esairlab.com/
[2] https://github.com/syehsyoh-dokling/filament-solution

Kaggle Competition, 2026, kaggle.com/competitions/filament-segmentation-2026
2026.

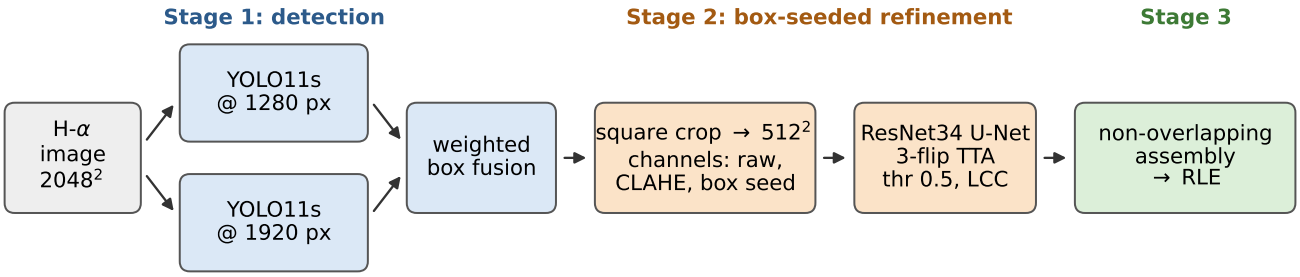


Figure 1: End-to-end pipeline. Stage 1 proposes boxes with two detectors and fuses them; Stage 2 segments one filament per box from a crop plus a box "seed"; Stage 3 resolves overlaps and writes COCO RLE.

within 2 px, median difference 0.5 px); the spine and chirality fields are never used. We split by *observation* (seed 42, 15%), so all passes of an image fall on the same side: 601 observations for training, 106 (170 annotator passes, 1,258 filaments) held out. All three models were trained on the same split.

The detectors use the Ultralytics default augmentation (40 epochs at 1280 px and 60 epochs at 1920 px). The refiner is trained on one crop per annotated filament for 12 epochs (AdamW, lr $3 \cdot 10^{-4}$, cosine schedule, loss 0.45 BCE + 0.55 soft Dice). To make the training seed look like a detector box, the ground-truth box is randomly shifted (±15%), rescaled (0.85–1.25) and padded (30–90%), with random flips. Checkpoints were selected on held-out Dice/mAP and the score threshold on held-out PQ, so held-out numbers are mildly optimistic; the public leaderboard is the unbiased estimate.

2.3 Examples

Figure 2 shows two held-out observations that each have two annotator passes, next to our prediction. The two annotators disagree on about half of the filaments; our prediction has as many instances as the more conservative annotator.

3 EVALUATION

3.1 Strategy

We score held-out predictions with the organizers' PQ definition [7]: a pair is a true positive when IoU >0.5, ground truth is grouped per annotator pass, and predictions per observation. Our implementation follows the organizers' self-evaluation notebook but works on RLEs. Uncertainty is a 95% bootstrap interval over the 106 held-out observations [3] (2,000 resamples; paired when comparing variants) and a Wilson interval for recalls [14]. For the three pipelines we scored both ways, held-out PQ ranked them in the same order as the public leaderboard, 0.04–0.06 higher.

Our leaderboard entries. Our team's highest public score (0.55), four further entries at 0.54–0.55 and one blend at 0.45 are *not* this method: they resubmitted, with minor post-processing, the output of a public notebook whose predictions are stored precomputed inside the notebook, which we could neither reproduce nor attribute to a model. We report only results our repository reproduces, and select the 0.39 entry as our final submission. Two of our submissions were rejected by the evaluator because instances overlapped; Stage 3 therefore enforces disjoint masks.

Table 1: Held-out PQ (95% CI) and public leaderboard (LB). Upper part: the path to the final pipeline; lower part: variants we tried and did not adopt (best score threshold for each).

Variant	PQ	TP/FP/FN	LB
Public notebook [12]	0.409 [.384,.434]	752/476/506	0.37
Ours, 1280 det., nearest upsampling	0.423	707/297/551	–
+ flip TTA, linear upsampling, LCC	0.437	720/288/538	0.38†
+ WBF with 1920 det. (**final**)	**0.444** [.419,.467]	750/326/508	**0.39**
1920 detector alone	0.433	746/374/512	
YOLO11m, 120 epochs	0.423	741/403/517	
YOLO11s, 120 epochs, new seed	0.427	755/418/503	
WBF of four detectors	0.441	732/311/526	
Detector test-time augmentation	0.432	757/391/501	
Refiner with clDice loss [10]	0.437	746/335/512	
Masks dilated by 1 px‡	0.392	734/493/524	
Masks eroded by 2 px‡	0.230	455/705/803	

† submitted at score threshold 0.28 (held-out 0.436). ‡ applied to the public-notebook predictions.

3.2 Results

Table 1 summarizes the measurements. The final pipeline improves on the public reference by +0.035 PQ (paired 95% CI +0.022 to +0.047). The step from one to two fused detectors is +0.008 and *not* significant (CI −0.001 to +0.018); we keep it because it was the better variant at every score threshold from 0.24 upward. The free inference changes in Stage 2 (flip TTA, linear instead of nearest upsampling, largest component) gave +0.014, mostly from the upsampling. Larger or longer-trained detectors, detector TTA, four-detector fusion and a skeleton-aware loss did not help. Three later attempts gave the same picture: adding a detector trained on all annotator passes (held-out 0.448, LB 0.39), retraining all models on 100% of the data (LB 0.38 alone, 0.39 in an ensemble with the original models), and re-ranking candidates with a classifier on 30 mask, contrast and position features (cross-validated 0.435–0.440).

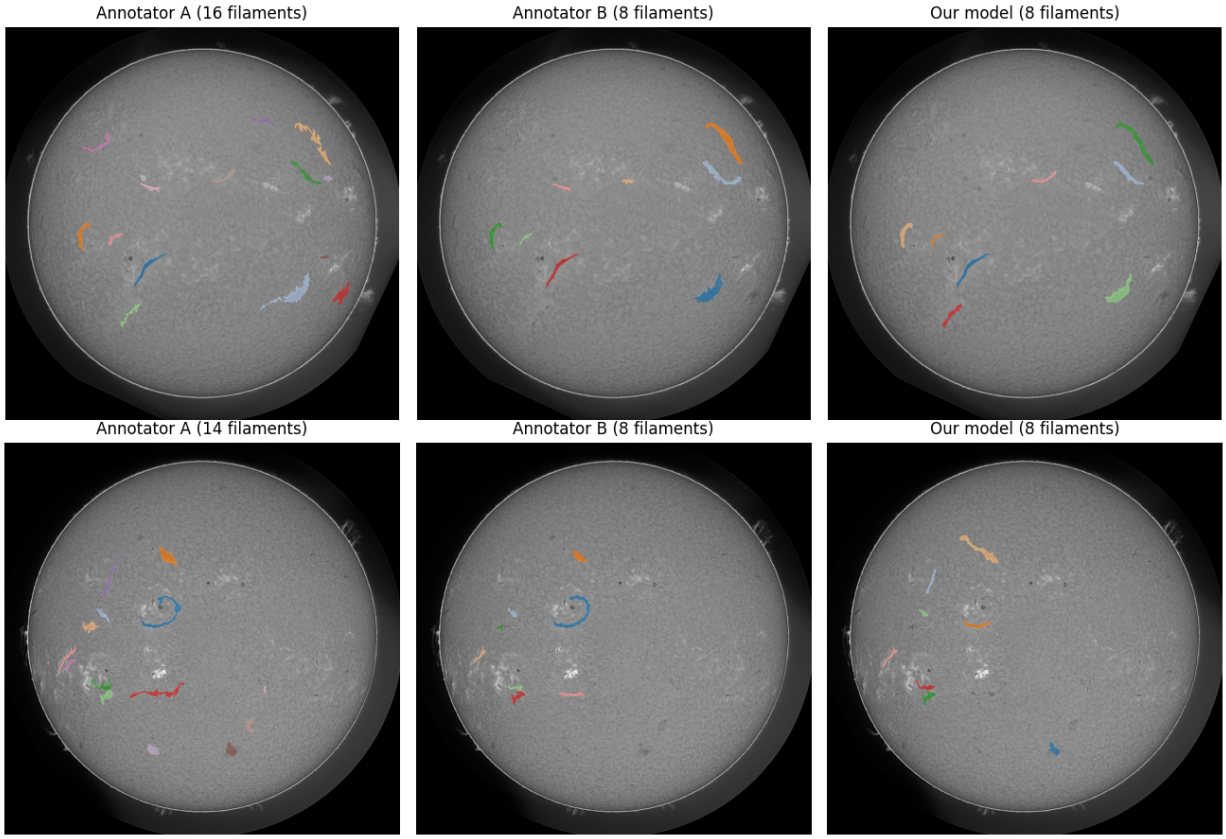


Figure 2: Two held-out observations, each annotated by two experts (left, middle), and our prediction (right). In both, annotator A marks 14–16 filaments and annotator B only 8.

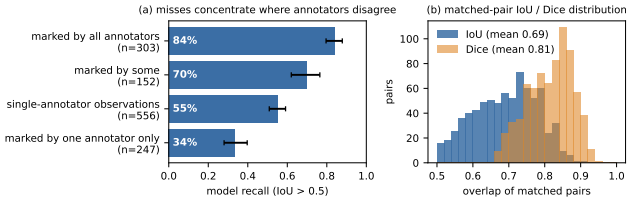


Figure 3: (a) Recall of the final pipeline on held-out filaments, grouped by how many annotators of the same observation marked them (95% Wilson intervals). (b) IoU and Dice of matched pairs.

At the image level, with all instances merged into one mask per annotator pass, the mean Dice (`torchmetrics DiceScore`) over the 170 held-out passes is 0.66 (95% CI 0.63–0.69; median 0.70, quartiles 0.59–0.77). Matched pairs have a mean IoU of 0.69 and a mean Dice of 0.81 (Fig. 3b). Counting every overlap (IoU >0) as in the organizers' notebook, 855 ground-truth filaments overlap exactly one prediction, 78 overlap several (fragmentation), and 325 none; 952 predictions overlap one filament, 35 several (over-merging), and 89 none. Over-merging and fragmentation are therefore minor; missed filaments dominate.

3.3 Where the errors are: annotator disagreement

Forty of the 106 held-out observations have two or more annotator passes. We scored each annotator's masks, as if they were predictions, against every *other* annotator of the same observation, and scored the model against exactly the same reference passes. Annotators reach PQ 0.34 (95% CI 0.30–0.38, mean matched IoU 0.63); the model reaches 0.45 (0.41–0.49, IoU 0.69). The paired difference is +0.11 (+0.08 to +0.14), and the model scores higher on 35 of the 40 observations. We read this as the model having learned the annotators' *common* convention, so it is closer to each of them than they are to one another, not as the model seeing filaments better than experts.

We grouped every ground-truth filament on these observations by how many of the *other* annotators also marked it (IoU >0.5). Figure 3a shows the result: the model finds 84% of the filaments marked by all annotators, 70% of those marked by some, and only 34% of those marked by a single annotator; the last group is also smaller (median 1,157 vs 1,663 px). On the 66 observations with a single annotator, where agreement is unknown, recall is 55%, between the two extremes. This is not a size effect: within each third

of the filament-area distribution, recall is 78–87% on consensus filaments and 31–36% on single-annotator ones, and in a logistic regression on both, the share of annotators who marked a filament is the dominant predictor ($z = 11.1$) while log-area is not significant ($z = 1.4$). Conversely, of the 213 predictions that count as false positives against one annotator, 112 (53%) were drawn by another annotator of the same image.

To separate detection from mask quality we fed the refiner the ground-truth boxes instead of detector boxes. PQ rises from 0.44 to 0.60, and on single-annotator observations the refiner then recovers 545 of 556 filaments with a mean IoU of 0.72, above the 0.63 inter-annotator IoU (the refiner is given the annotator's own box here, which favours it). Mask shape is thus close to human agreement; the remaining gap is about *which* structures count as filaments, a decision on which the experts themselves differ.

3.4 Strengths, weaknesses and improvements

Strengths. The pipeline is fast and deterministic, its instances never overlap, and it reproduces its submission byte-for-byte; on consensus filaments its recall is 84%. **Weaknesses.** Recall on filaments that only one annotator marks is low (34%), whatever their size; masks are sensitive to their boundary (a 1 px dilation already costs 0.017 PQ, a 2 px erosion 0.18), which also explains why our earlier global 512 px semantic model scored only 0.29 on the leaderboard; one test observation receives no instance because the two stages produce no non-empty candidate for it. **Improvements.** Training on consensus-weighted targets (e.g., soft labels from the number of annotators who marked a filament) instead of treating every pass as equally correct (pooling all passes as detector targets was not enough), and calibrating the detector's confidence to annotator agreement. For evaluation we suggest reporting recall stratified by annotator agreement next to PQ, and PQ relative to the inter-annotator ceiling; both are cheap to compute and make scores comparable across datasets with different labelling conventions.

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